# Course: 405-05: Applied Artificial Intelligence with Open-Source Development Tools

| | |
|---|---|
| **Course Code** | 405-05 |
| **Course Title** | Applied Artificial Intelligence with Open-Source Development Tools |
| **Credit** | 4 |
| **Course Category:** | Major Course |
| **Level of Course:** | 300- 399 ( Higher Course ) |
| **Teaching per Week** | 2 Hours Theory + 4 Hours of Lab. Work |
| **Minimum weeks per Semester** | 15 (Including class work, examination, preparation etc.) |
| **Review / Revision** | - |
| **Implementation Year:** | 2025-2026 |
| **Purpose of Course** | The purpose of this course is to equip students with hands-on, practical knowledge in the field of Artificial Intelligence using widely adopted open-source tools and frameworks. It aims to bridge the gap between theoretical AI concepts and real-world applications by enabling learners to design, develop, and deploy AI-based solutions using Python, machine learning libraries, chatbot frameworks, and data visualization platforms. The course emphasizes skill-building through practical labs, case studies, and mini-projects focused on automation, natural language interaction, and data-driven decision-making. |
| **Course Objective** | OB1: Understand the role and application of open-source AI tools in real-world domains.<br>OB2: Develop and deploy machine learning models using Python-based tools.<br>OB3: Analyze and process natural language data using NLP libraries.<br>OB4: Design, build, and deploy rule-based and AI-based chatbots using open-source platforms.<br>OB5: Apply practical knowledge through mini-projects involving real datasets and chatbot use-cases. |
| **Pre-requisite** | Basic understanding of programming using Python, including data types, functions, and control structures. Familiarity with foundational concepts of machine learning, supervised/unsupervised algorithms, and experience using Python libraries such as NumPy, Pandas, and Matplotlib is essential. |
| **Course outcome** | **CO1:** Identify and explain open-source tools used in AI applications. (Remembering, Understanding)<br>**CO2:** Implement AI models using Python libraries and visualize results using dashboards. (Applying)<br>**CO3:** Analyze natural language data using open-source NLP frameworks. (Analyzing)<br>**CO4:** Design and build rule-based and AI-based chatbots using Python and Rasa. (Creating)<br>**CO5:** Evaluate the chatbot model performance and enhance interaction through testing and feedback. (Evaluating) |
| **Mapping between Course Outcome(CO) and Program Specific Outcome (PSO):** | |

| | PSO1 | PSO2 | PSO3 | PSO4 | PSO5 | PSO6 | PSO7 | PSO8 |
|---|---|---|---|---|---|---|---|---|
| CO1 | | | | | | | | |
| CO2 | | | | | | | | |
| CO3 | | | | | | | | |
| CO4 | | | | | | | | |
| CO5 | | | | | | | | |

| | |
|---|---|
| **Course Content** | **UNIT 1: Introduction to Applied Artificial Intelligence and Open-Source Tools**<br>1.1 Understanding Applied AI<br>1.1.1 Difference between Theoretical AI vs. Applied AI<br>1.1.2 Use Cases of AI in Industries (Healthcare, Education, Finance, Marketing)<br>1.2 Overview of Open-Source AI Tools<br>1.2.1 What is Open Source in AI?<br>1.2.2 Advantages and Limitations of Open-Source Tools<br>1.2.3 Community, Licensing, and Support<br>1.3 Introduction to AI Tools for Data Preparation and Modeling<br>1.3.1 Google Colab and Jupyter Notebook<br>1.3.2 Scikit-learn for Machine Learning<br>1.3.3 OpenAI Gym for Reinforcement Learning Basics<br>1.3.4 Introduction to Hugging Face Transformers<br>1.4 Case Studies<br>1.4.1 AI in Retail: Predicting Customer Purchase<br>1.4.2 AI in Healthcare: Disease Diagnosis System<br><br>**UNIT 2: Practical Exploration of Open-Source AI Tools**<br>2.1 Natural Language Processing Tools<br>2.1.1 SpaCy and NLTK: Text Cleaning and Tokenization<br>2.1.2 Sentiment Analysis using TextBlob<br>2.1.3 Named Entity Recognition with SpaCy<br>2.2 AI Model Development Tools<br>2.2.1 TensorFlow and Keras: Building a Simple Neural Network<br>2.2.2 AutoML Tools: H2O.ai and MLJAR<br>2.2.3 MLFlow for Model Tracking and Experimentation<br>2.3 Open-Source AI Visualization Tools<br>2.3.1 Streamlit: Creating AI-powered Web Apps<br>2.3.2 Gradio: Deploying AI Models with GUI<br>[ Recommended Lab : (i) Build and deploy a Sentiment Analyzer (ii) Create an AI-powered dashboard using Streamlit ]<br><br>**UNIT 3: Introduction to Chatbots and Their Applications**<br>3.1 Understanding Chatbot<br>3.1.1 Types of Chatbots: Rule-Based and AI-Based<br>3.1.2 Chatbot Architecture<br>3.1.3 Chatbots in Business and Customer Support<br>3.2 Designing Chatbot Workflows<br>3.2.1 Defining User Intents and Responses<br>3.2.2 Creating Dialog Flows<br>3.2.3 Handling FAQs and User Context<br>3.3 Tools for Chatbot Development<br>3.3.1 Rasa: Open-Source AI Chatbot Framework<br>3.3.2 ChatterBot: Rule-based Chatbots in Python<br>3.3.3 Dialogflow (overview and comparison with open-source tools)<br>3.4 Real-Life Examples and Use Cases using Rasa or ChatterBot<br>3.4.1 Chatbots in E-commerce<br>3.4.2 Educational Bots<br>3.4.3 Healthcare Assistants<br><br>**UNIT 4: Practical Chatbot Development Using Python**<br>4.1 Building a Rule-Based Chatbot using ChatterBot<br>4.1.1 Using ChatterBot with Python<br>4.1.2 Training Chatbot with Corpus Data<br>4.1.3 Deploying on Localhost or Streamlit<br>4.2 AI-Based Chatbot with Rasa<br>4.2.1 Installation and Configuration of Rasa<br>4.2.2 Defining Domain, NLU, and Stories<br>4.2.3 Creating Custom Actions and Webhooks |

| | |
|---|---|
| | 4.2.4 Deploying Rasa Chatbot in Web Application<br>4.3 Integration and Testing<br>4.3.1 Integrating Chatbot with Web UI (Rasa UI or HTML + Flask for frontend)<br>4.3.2 Testing and Improving UX (Rasa shell NLU, conversation tests)<br>4.3.3 Logging and Analytics (Rasa supports conversation trackers and logging)<br>4.4 Developing Mini Project: End-to-End Chatbot Development<br>4.4.1 Use cases (Helpdesk Bot / Health Assistant / Feedback Bot)<br>4.4.2 Design, Build, Train, and Deploy Chatbot<br>4.4.3 Project with Live Demo and Report |
| **Lab Exercises:** | Lab 1: Building a Rule-Based Chatbot with ChatterBot<br>Objective: Create a simple rule-based chatbot using Python.<br>Tasks:<br>Install ChatterBot and dependencies.<br>Create a chatbot instance.<br>Train chatbot with ChatterBot's English corpus data.<br>Test chatbot responses with sample queries.<br>Deploy chatbot on localhost using Streamlit interface.<br>Deliverable: Functional chatbot answering basic questions.<br>Lab 2: Custom Training and Improving ChatterBot Responses<br>Objective: Customize chatbot training for specific domain.<br>Tasks:<br>Prepare a small custom corpus related to a chosen domain (e.g., library assistant).<br>Train ChatterBot with custom corpus.<br>Evaluate chatbot's response accuracy.<br>Implement conversation logging to analyze interactions.<br>Deliverable: Domain-specific chatbot with improved response accuracy.<br>Lab 3: Installing and Setting Up Rasa Environment<br>Objective: Set up Rasa on local machine.<br>Tasks:<br>Install Rasa and dependencies.<br>Initialize a new Rasa project.<br>Explore folder structure and key files (domain.yml, config.yml, data folder).<br>Deliverable: Running Rasa project ready for development.<br>Lab 4: Defining NLU Components and Intents in Rasa<br>Objective: Build intent recognition model.<br>Tasks:<br>Define intents and example utterances in nlu.yml.<br>Train NLU model using Rasa CLI.<br>Test intent classification using Rasa shell.<br>Deliverable: NLU model classifying user intents accurately.<br>Lab 5: Creating Domain and Stories for Dialogue Management<br>Objective: Manage conversations using stories.<br>Tasks:<br>Define domain entities, slots, responses in domain.yml.<br>Create stories to map intents to responses.<br>Train dialogue management model.<br>Test chatbot conversation flow using Rasa shell.<br>Deliverable: Chatbot capable of multi-turn conversations.<br>Lab 6: Developing Custom Actions and Webhook Integration<br>Objective: Extend chatbot functionality using custom actions.<br>Tasks:<br>Write Python code for custom actions (actions.py).<br>Connect custom actions via endpoints.yml.<br>Test custom actions locally.<br>Deliverable: Chatbot performing backend tasks (e.g., API calls).<br>Lab 7: Deploying Rasa Chatbot as a Web Application<br>Objective: Integrate chatbot with web interface.<br>Tasks:<br>Build simple HTML/JavaScript frontend to interact with Rasa chatbot.<br>Deploy Rasa chatbot backend using Flask or FastAPI. |

| | |
|---|---|
| | Test end-to-end communication between frontend and chatbot.<br>Deliverable: Fully deployed chatbot accessible via web browser.<br>Lab 8: Logging and Analytics of Chatbot Interactions<br>Objective: Monitor chatbot usage and performance.<br>Tasks:<br>Enable conversation trackers in Rasa.<br>Collect logs of user interactions.<br>Analyze logs for common queries, errors, fallback usage.<br>Deliverable: Analytical report on chatbot performance.<br>Lab 9: Mini Project – End-to-End Chatbot Development<br>Objective: Apply learned skills to build a complete chatbot.<br>Tasks:<br>Select a use case (e.g., Helpdesk Bot, Health Assistant, Feedback Bot).<br>Design intents, domain, and stories.<br>Develop custom actions as needed.<br>Train and test chatbot.<br>Deploy chatbot locally or on cloud.<br>Prepare project report and demo presentation.<br>Deliverable: Functional chatbot with full documentation and live demo. |
| **Reference Books** | 1) Artificial Intelligence: A Guide to Intelligent Systems, Michael Negnevitsky, Pearson Education, ISBN: 9788131708430<br>2) Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, Aurélien Géron, O'Reilly, ISBN: 9781098125974<br>3) Natural Language Processing with Python, Steven Bird, Ewan Klein, O'Reilly, ISBN: 9780596516499<br>4) Building Chatbots with Python, Sumit Raj, Apress, ISBN: 9781484240958<br>5) Conversational AI: Chatbots that Work, Michael McTear, Springer, ISBN: 9783030621545<br>6) Machine Learning with Python, Abhishek Vijayvargia, BPB Publications, ISBN: 9789388176623<br>7) Deep Learning with Python, François Chollet, Manning Publications, ISBN: 9781617294433<br>8) Designing Bots, Amir Shevat, O’Reilly, ISBN: 9781491974827<br>9) Mastering Rasa, Greg Stephens, Packt Publishing, ISBN: 9781800562545<br>10) Streamlit for Data Science, Tyler Richards, O’Reilly (online)<br>11) Documentation of Rasa, https://rasa.com/docs<br>12) Hugging Face Tutorials, https://huggingface.co/learn |
| **Teaching Methodology** | Class Work, Discussion, Self-Study, Seminars and/or Assignments |
| **Evaluation Method** | 50% Internal assessment.<br>50% External assessment. |